
FINE-GRAINED VISUAL KNOWLEDGE IS A DISTINCT CAPABILITY OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have advanced rapidly on visual question answering benchmarks spanning visual reasoning, document understanding, and multimodal dialogue. These gains now appear across diverse base models, alignment architectures, and training data, but they do not guarantee strong fine-grained visual knowledge: recent evidence shows that VLMs still lag on traditional image classification benchmarks that require precise category discrimination. We provide a broad evaluation of recent VLMs on fine-grained classification and directly isolate the factors behind the gap between fine-grained knowledge and standard vision-language benchmarks. Across a targeted ablation suite, stronger LLMs improve all benchmark scores at similar rates, whereas stronger vision encoders deliver disproportionate gains on fine-grained classification. We further show that pretraining is a central driver of fine-grained performance, especially when language model weights are unfrozen during pretraining. These findings establish concrete design priorities for building VLMs with stronger fine-grained visual understanding and more vision-centric capabilities.

1 INTRODUCTION

Vision-language models (VLMs), which connect vision encoders to large language models (LLMs), now demonstrate sophisticated abilities to interpret, describe, and reason over visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). They handle demanding tasks such as open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). Yet their progress leaves a central capability unresolved: whether modern VLMs possess the fine-grained visual perception needed to distinguish closely related categories, and which design choices most reliably improve that capability.

Fine-grained visual recognition—the ability to distinguish between visually similar categories—requires models to attend to subtle discriminative features while remaining robust to intra-class variation in pose, lighting, and background. Traditional vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have established strong performance on fine-grained classification benchmarks, but modern VLMs have not been examined with the same rigor on this fundamental visual skill.

Fine-grained visual knowledge is not a peripheral benchmark property; it is a prerequisite for reliable visual reasoning. Real-world VLM deployments routinely face long-tailed distributions where small category differences determine the correct downstream action. If a model misidentifies a mushroom species, for example, its textual knowledge about toxicity cannot support a safe consumption decision. Pinpointing how VLMs acquire or lose fine-grained knowledge therefore provides actionable guidance for architectures and training recipes that must support precise visual understanding.

We present a comprehensive evaluation of fine-grained visual knowledge in state-of-the-art VLMs and contrast it with performance on general VLM tasks. Specifically, we evaluate 15 VLMs, including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024), on four fine-grained classification benchmarks—ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014)—alongside aggregated results from eight general VLM benchmarks, including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). The evaluation exposes large

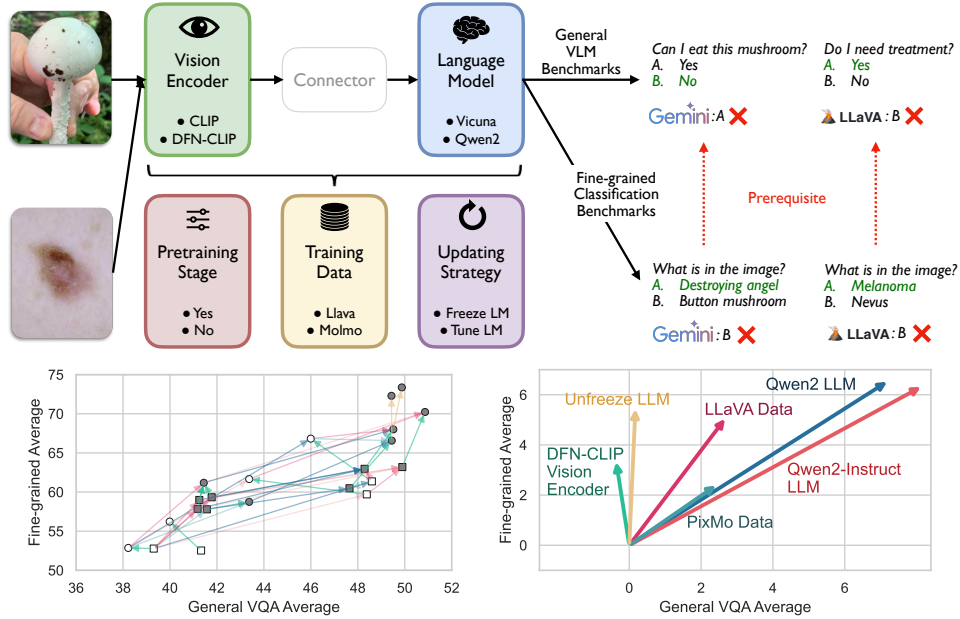


Figure 1: **Overview.** (Top) We evaluate fine-grained classification as a core vision-language capability that current benchmark practice underemphasizes despite its role in higher-level understanding and reasoning. (Bottom) Through 22 systematic ablation experiments over model components and training strategies (left), we identify the design factors that decisively drive fine-grained classification performance in VLMs (right).

fine-grained performance differences among models with comparable general abilities, establishing fine-grained recognition as a distinct axis of visual intelligence that standard VLM benchmarks do not capture. We also confirm a persistent gap between VLMs and their CLIP-based vision encoders, consistent with prior findings (Zhang et al., 2024; Geigle et al., 2024).

To determine what actually drives fine-grained classification performance, we conduct a systematic ablation study over core model components and training paradigms. We isolate the impact of language models, vision encoders, pretraining, data quality, and weight-updating choices. Across 22 carefully controlled ablations, we identify the factors that govern fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Stronger base language models raise performance consistently across both fine-grained classification and general VLM benchmarks.
- **Vision encoder (§4.2):** Stronger vision encoders deliver targeted gains in fine-grained classification while producing more limited changes on general VLM benchmarks, especially in two-stage trained models.
- **Pretraining stage (§4.3):** Large-scale image-caption pretraining is a major driver of fine-grained classification performance and has a smaller effect on general VLM benchmarks.
- **Weight updating method (§4.3):** Fine-grained classification improves when pretraining updates both the connector and the LLM, rather than the connector alone.
- **Data quality (§4.3):** Within our tested settings, pretraining and instruction-tuning data quality has a limited effect on fine-grained classification, particularly when language model weights remain frozen during pretraining.

Overall, this work positions fine-grained visual perception as an essential and currently undermeasured dimension of VLM evaluation. Our ablations convert that observation into concrete design guidance: base LLMs, vision encoders, pretraining scale, and LLM updating play distinct roles in fine-grained knowledge acquisition. The resulting evidence provides a clear path toward more vision-centric VLMs that are better suited to real-world long-tail tasks requiring precise recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) integrate visual inputs with language models and often follow the LLaVA-style architecture (Liu et al., 2023): a vision encoder, a language model, and an MLP connector that aligns the two. This design has produced strong performance across multimodal tasks including visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). The field has responded with broad VLM benchmarks such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). These benchmarks, however, primarily measure reasoning and language use conditioned on images; they leave core vision-centric abilities such as object recognition and fine-grained classification undermeasured. Our work directly closes this evaluation gap by systematically measuring and analyzing fine-grained classification, an essential component of visual intelligence.

Fine-grained visual classification. Fine-grained classification is a mature computer vision task that tests whether models can distinguish visually similar subcategories within broader object classes, such as bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). These benchmarks have been central to developing and validating vision encoders including ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Despite their established importance in vision research, recent VLM evaluations largely replace these datasets with visual question-answering benchmarks. Because real applications often involve long-tail distributions that demand precise recognition—including medical diagnosis, food safety, and species identification—fine-grained classification is necessary for assessing whether VLMs are practically usable as vision-centric systems rather than only language-centric reasoners.

Evaluation and ablation of VLMs for fine-grained classification. Recent studies have begun evaluating VLMs on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), but their scope remains narrow, typically covering only a small set of models or datasets. We substantially extend this line of work by evaluating a broader set of VLMs across multiple fine-grained classification benchmarks under a unified multiple-choice protocol. More importantly, we move beyond measurement alone: our ablation analysis identifies the model and training factors that determine fine-grained classification performance.

Several recent works map the broader VLM design space, studying architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and multimodal dataset composition (Gadre et al., 2023; Udandara et al., 2024; Deitke et al., 2024). Those analyses mainly optimize and interpret performance on general benchmarks that emphasize language and reasoning. In contrast, we isolate how base model selection, pretraining methods, and pretraining or instruction-tuning data composition affect fine-grained classification, yielding specific evidence for optimizing VLMs on vision-centric tasks.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section motivates fine-grained classification as a necessary VLM evaluation axis, defines the benchmark protocol, and reports the central empirical patterns from 15 existing VLMs.

3.1 MOTIVATION

Fine-grained classification is a foundational computer vision task that requires distinguishing visually similar subcategories within a broader class, such as bird species or car models (Wei et al., 2021). Benchmarks such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) have therefore served as standard testbeds for training and evaluating vision encoders. With large-scale self-supervised and supervised learning, vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have reached near- or superhuman performance on challenging fine-grained classification tasks.

VLM evaluation has not kept this vision-centric axis at the center. Although recent VLMs such as LLaVA (Liu et al., 2023) show strong abilities to interpret visual concepts, they are typically evaluated through visual question answering (VQA), including academic VQA benchmarks such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document understanding tasks

such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). These benchmarks emphasize reasoning and language processing over fine-grained object categories, leaving a core visual capability largely untested.

Fine-grained classification is therefore critical for VLMs: real-world applications frequently involve long-tailed categories where precise visual knowledge separates correct from unsafe decisions. If a blind user asks whether a mushroom is safe to eat, for example, the model must first identify the species before using textual knowledge about toxicity; misclassifying a deadly destroying angel as an edible white button mushroom could have fatal consequences. The same requirement appears in medical diagnosis, where visually similar diseases can imply different treatments, and in autonomous driving, where confusing a “stop” sign with a “do not enter” sign could trigger dangerous navigation behavior. Reliable VLMs must therefore perform well on fine-grained classification, not merely on broad visual reasoning benchmarks.

3.2 BENCHMARKS

We evaluate VLM fine-grained visual knowledge on four established object-recognition benchmarks that cover distinct fine-grained domains: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

These benchmarks contain 37 to 1,000 classes, which makes direct VLM evaluation in the multiple-choice format common to VLM training and evaluation nontrivial. We therefore adopt the method of Geigle et al. (2024) and convert each dataset into 5-way multiple-choice questions. For each image, an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) selects the four hardest negative choices by image-label cosine similarity, which we combine with the ground-truth label to form the answer set. We use OpenCLIP specifically to avoid biasing difficulty against VLMs that rely on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), all of which are trained on datasets different from the OpenCLIP model. This conversion tests VLMs in their native evaluation format while retaining substantial classification difficulty. For example, CLIP ViT-L/14-336px (Radford et al., 2021) obtains 75.2% accuracy on the 102-class Flowers-102 benchmark and only rises to 78.3% accuracy on our 5-way multiple-choice version.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abdin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM, we use its default VLMEvalKit prompt format (Duan et al., 2024) and score accuracy by exact match to the multiple-choice options. For a vision-encoder baseline, we evaluate CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) in the standard zero-shot setup, encoding images and class labels (“A photo of a [class]”) with CLIP image and text encoders. We compute cosine similarities between image and text embeddings and take the highest-scoring class as the top-1 prediction.

To test whether standard VLM benchmarks already capture fine-grained classification ability, we compare fine-grained classification accuracy with each model’s average score on eight general VQA benchmarks from VLMEvalKit, including MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024).

3.4 RESULTS

Figure 2 plots fine-grained classification performance, averaged across our four benchmarks, against general VQA performance, averaged across eight existing VQA benchmarks, for every tested VLM. The relationship makes the separation clear: models with similar general VQA scores can differ dramatically in fine-grained classification accuracy. For example, CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) both achieve nearly 48% average accuracy on general VQA benchmarks, yet their fine-grained classification accuracies differ by 19pp (percentage points)—77.3% for CogVLM-Chat versus 58.1% for LLaVA-NeXT-Vicuna-13B. This discrepancy

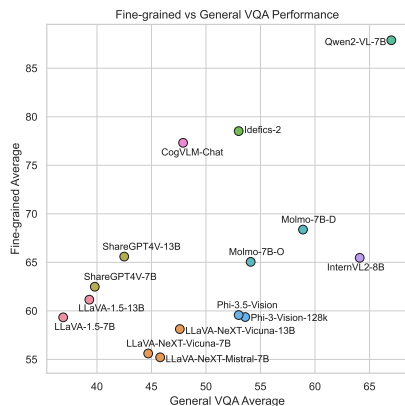


Figure 2: **Fine-grained classification compared to general VQA performance across VLMs.** Recent VLMs separate sharply along this axis, showing that fine-grained classification is a distinct visual capability that standard VQA benchmarks do not measure.

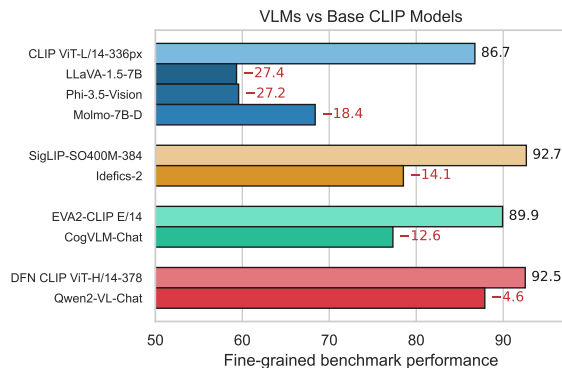


Figure 3: **Comparison of VLMs with their corresponding CLIP vision encoders in fine-grained classification.** Qwen2-VL-Chat nearly matches its DFN-CLIP vision encoder, but the other evaluated VLMs remain far behind their corresponding encoders, revealing a substantial opportunity to preserve and exploit encoder-level fine-grained knowledge.

demonstrates that existing VLM evaluation benchmarks do not adequately capture fine-grained classification, establishing it as a distinct dimension of visual intelligence.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Current VLMs typically inherit their visual representations from pretrained vision encoders, which lets us compare VLMs directly with the corresponding CLIP baselines under our multiple-choice protocol. Figure 3 shows that the VLM interface often fails to retain the fine-grained classification strength already present in the encoder. Molmo (Deitke et al., 2024), a recent model with strong VQA performance, achieves 68.4% average accuracy in our multiple-choice evaluation, whereas its CLIP ViT-L/14 vision encoder (Radford et al., 2021) reaches 86.7%. Even the strongest tested model, Qwen2-VL-Chat (Wang et al., 2024b), remains 4.6 points below its own state-of-the-art DFN-CLIP vision encoder (Fang et al., 2023). This consistent gap motivates the ablation study in the next section, where we identify the design choices that recover and improve fine-grained classification capability.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the observational evidence in Section 3, we ablate the model and training choices that plausibly drive fine-grained classification performance, with a primary focus on base model selection and training methodology. We use the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023) as a controlled substrate. Each model receives one epoch of pretraining when applicable, followed by one epoch of finetuning. Across experiments, we retain LLaVA-1.5’s default hyperparameters, including learning rate and batch size, so that performance changes reflect the ablated factors rather than incidental tuning differences.

We evaluate fine-grained performance as the average score across the four classification benchmarks described in Section 3.2. For general VQA performance, we report the average score on three widely used multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and

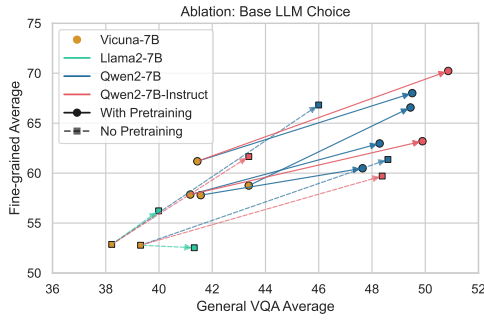


Figure 4: Ablating the base LLM by replacing Vicuna-7B in LLaVA. Switching from Vicuna to Qwen2-7B produces broad gains: +7.5pp in fine-grained performance and +7.7pp in general VQA performance on average.

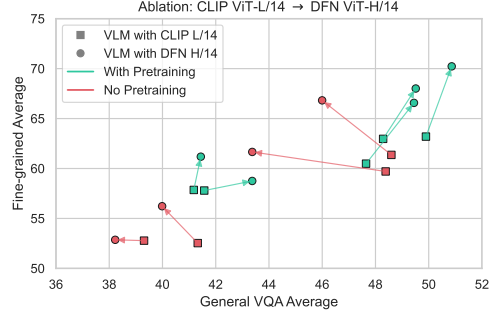


Figure 5: Ablating the base vision encoder by replacing CLIP ViT-L/14 in LLaVA with DFN-CLIP ViT-H/14. With sufficient pretraining, DFN-CLIP yields a targeted +4.5pp gain in fine-grained performance and a smaller +1.2pp gain in general performance.

MMStar (Chen et al., 2024b). These benchmarks provide broad domain coverage and are widely adopted in VLM evaluation, making them a strong contrast set for fine-grained classification.

4.2 BASE MODELS

Section 3 shows that recent models often differ in their base LLMs while many still rely on the CLIP ViT-L/14 vision encoder from Radford et al. (2021). We therefore isolate how these base model choices shape fine-grained classification rather than treating them as incidental architectural details.

4.2.1 LLM CHOICE

Setup. Using LLaVA-1.5-7B (Liu et al., 2023) as the baseline, we ablate the LLM component across vision encoders and pretraining configurations. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023) used in LLaVA against Qwen2-7B (Yang et al., 2024), which underlies Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024), as well as Llama-2-7B (Touvron et al., 2023), Vicuna’s base model, and Qwen2-7B-Instruct, the instruction-tuned Qwen2-7B variant.

Findings. Figure 4 shows a consistent result: replacing Vicuna-7B with Qwen2-7B substantially improves every configuration, with average gains of +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. Qwen2-7B-Instruct delivers nearly identical gains (+7.5pp and +8.1pp, respectively). Llama2-7B, the non-instruction-tuned base of Vicuna, provides smaller gains, although the downstream implications of removing instruction tuning remain unresolved. Importantly, all trained models, including the weaker Vicuna-based VLMs, reliably produce multiple-choice responses in the correct format. Thus, the observed performance differences reflect better knowledge and option discrimination rather than improved format adherence.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most evaluated VLMs, including the LLaVA series and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as the vision encoder. Qwen2-VL (Wang et al., 2024b), in contrast, uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which has stronger zero-shot object-recognition performance on tasks such as ImageNet. We compare these two encoders across LLM choices and pretraining configurations to isolate the contribution of visual representation quality.

Findings. Figure 5 demonstrates that the benefit of replacing CLIP L/14 with DFN-CLIP H/14 depends on whether the connector is pretrained before finetuning. Without pretraining, DFN-CLIP

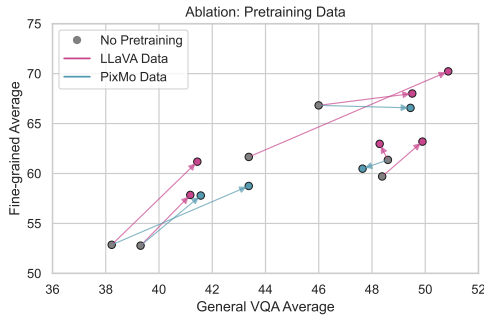


Figure 6: **Ablating pretraining data.** Adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) data produces clear gains. LLaVA data raises fine-grained and general scores by +4.0pp and +2.1pp, respectively, while PixMo data raises them by +2.4pp and +2.5pp.

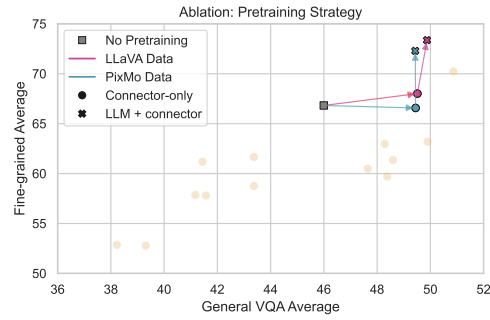


Figure 7: **Ablating training method by comparing connector-only pretraining with LLM+connector tuning.** Unfreezing the LLM during pretraining delivers a +5.5pp fine-grained gain without degrading general VQA scores.

Table 1: **Summary of important ablation settings and comparison to existing models (bolded).** The ablations show that stronger base LLMs, stronger vision encoders, and pretraining that updates both the connector and LLM improve fine-grained performance, while instruction finetuning contributes less. General VQA follows a different pattern, with LLM choice accounting for a larger share of the gains.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

reduces general VQA performance (-2.5pp) while only modestly improving fine-grained scores (+2.8pp). After connector pretraining, however, general VQA performance stays stable or slightly improves (+1.2pp), while fine-grained performance rises substantially (+4.5pp). This result establishes that stronger vision encoders translate into fine-grained VLM gains when alignment training properly integrates them.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pre-training before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are typically trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). Pretraining uses large-scale image-caption data and a captioning objective to align the vision encoder with the LLM. Instruction tuning then refines the model on curated multi-round, multimodal instruction-completion data so that responses better match human intent in downstream applications. We analyze these stages directly because our results show that training choices, especially pretraining, strongly determine fine-grained capability.

4.3.1 PRETRAINING

Setup. We target three central questions about the pretraining process:

- Does pretraining remain necessary for LLaVA-architecture models when the base models are already pretrained on billion-scale data?
- How does connector-only pretraining compare with pretraining that also unfreezes the LLM?
- How much does pretraining data quality change downstream performance?

We compare three pretraining configurations: LLaVA pretraining data (Liu et al., 2023), a subset of CC-3M with web-scraped captions; Molmo pretraining data (Deitke et al., 2024), PixMo-Cap with detailed human-annotated captions; and no pretraining. In most experiments, we pretrain only the connector, following Liu et al. (2023). We also test a strategy similar to Deitke et al. (2024): pretrain the connector for the first 20% of steps, then unfreeze the LLM for the remaining 80%. Every configuration is then finetuned for one epoch on LLaVA finetuning data.

Findings. Figure 6 reports the pretraining ablations. Connector-only pretraining generally improves VLM performance, with the Qwen2 and CLIP ViT-L/14 combination as the exception. The gains are especially large for Vicuna-based VLMs, which improve by +6.1pp on fine-grained classification and +3.1pp on general VQA on average.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

Within this setup, caption quality is not the dominant factor: pretraining on lower-quality web-scraped captions (LLaVA) and on detailed human annotations (PixMo) yields similar outcomes. Compared with LLaVA data, PixMo data changes fine-grained performance by -1.6pp and general VQA by +0.4pp.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

We hypothesize that connector-only pretraining limits the model’s ability to exploit richer captions. To test this, we pretrain both the LLM and connector after a 20% connector-only warmup. Figure 7 shows that this strategy decisively improves fine-grained classification (+5.5pp) while preserving general VQA performance. The improvement appears with both LLaVA and PixMo pretraining data, indicating that the lower-quality captions do not measurably damage the LLM’s language capability in this setting.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. For this ablation, we start from the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b), the base model used to train Qwen2-VL-7B-Chat, and finetune it on LLaVA-Instruct data (Liu et al., 2023) using our earlier protocol. We compare the resulting model with Qwen2-VL-7B-Chat, which uses 2M in-house instruction-finetuning examples (Wang et al., 2024b). This isolates the contribution of the instruction-finetuning dataset after the same pretrained Qwen2-VL foundation.

Findings. Finetuning only on LLaVA data slightly lowers both fine-grained and general benchmark performance. Fine-grained scores decrease by 2.4pp, from 87.9% to 85.5%, while general benchmark scores decrease by 1.3pp, from 62.4% to 61.1%. As Figure 8 shows, this is the smallest fine-grained effect among our ablations, making instruction finetuning less central to fine-grained knowledge acquisition than base model selection and pretraining strategy.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

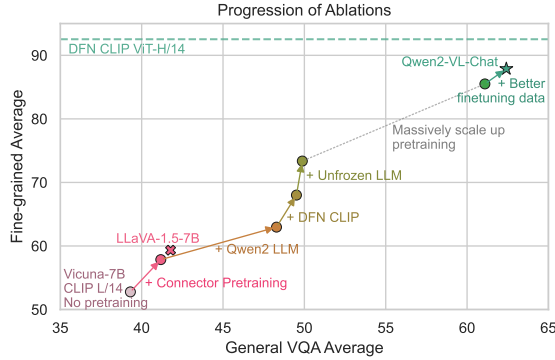


Figure 8: **Progression of combined ablation effects.** Starting from Vicuna-7B and CLIP ViT-L/14, successive changes to the base LLM, vision encoder, and training settings raise fine-grained classification accuracy from 52.8% to 73.4%, closing more than half of the gap to Qwen2-VL-Chat’s 87.9%. The remaining difference is most consistent with Qwen2-VL’s much larger 1.4T-token pretraining scale.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 synthesize the ablations and quantify how each component closes the gap between weaker and stronger VLMs on fine-grained and general VQA tasks. Starting from the baseline LLaVA architecture (Liu et al., 2023), with Vicuna-7B (Chiang et al., 2023) and CLIP ViT-L/14 (Radford et al., 2021), the combined modifications raise fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The pattern is clear: switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately strengthen fine-grained capabilities, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) explains most of the general VQA improvement.

Even after these substantial gains, a 12-point fine-grained classification gap remains relative to Qwen2-VL-Chat (Table 1). We examine two plausible sources of this remaining gap: architectural differences and pretraining data scale.

Architectural changes may contribute, but the observational results in Section 3.4 and the training ablations in Section 4.3 point more strongly to pretraining data scale. Our LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) pretraining runs use relatively small datasets, each with fewer than 1M images and captions, corresponding to approximately 200M and 400M tokens, respectively. By contrast, Wang et al. (2024b) report pretraining Qwen2-VL on 1.4T tokens, orders of magnitude beyond our controlled ablations. This scale disparity provides the most plausible explanation for the remaining gap and reinforces extensive pretraining as a critical ingredient for VLMs with strong fine-grained classification capability.

5 LIMITATIONS & CONCLUSION

Our study has practical limitations, but they also sharpen the next research priorities. Because of computational constraints, our controlled training comparisons use <1M data points rather than the billion-scale training regimes used by some newer VLMs; as a result, our experiments identify the direction and relative importance of key factors but do not fully characterize the effects of frontier-scale pretraining. Newer work also proposes alternative pretraining strategies, and those strategies may interact with fine-grained visual understanding in ways that warrant direct study.

This work establishes fine-grained visual classification as a crucial and undermeasured dimension of vision-language models (VLMs). Through a systematic evaluation of state-of-the-art VLMs and a controlled analysis of model components and training paradigms, we show that fine-grained performance depends strongly on vision encoder quality, pretraining, and whether the LLM is updated during pretraining, while base LLM improvements produce broader benchmark gains. These results provide concrete guidance for building more vision-centric VLMs and for deploying them in real-world settings where precise visual understanding is essential.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: **Detailed results of model ablations.** Aggregated scores show how each controlled design change shifts both fine-grained classification and general VQA performance.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: **Detailed results of model ablations on general VQA benchmarks.** The per-benchmark breakdown supports the main conclusion that LLM choice drives the largest general VQA gains.

A FULL RESULTS

We report the complete ablation results for aggregated performance, general VQA benchmarks, and fine-grained classification in Table 2, Table 3, and Table 4, respectively, providing the full evidence behind the trends emphasized in the main text.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Detailed results of model ablations on fine-grained classification benchmarks.** The per-dataset breakdown shows the targeted benefits of stronger vision encoders and LLM-updating pretraining for fine-grained recognition.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As described in Section 3.2, we evaluate VLMs and their base vision encoders on four established classification benchmarks that directly test fine-grained recognition:

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset organized around the WordNet hierarchy, with broad supercategories that contain many fine-grained subcategories. The ImageNet test set contains 50,000 images spanning 1,000 classes.

The ImageNet/WordNet hierarchy contains many classes that are too visually distinct from other classes to support hard multiple-choice distractors; for example, only three classes fall under the “person” category. To construct a challenging and tractable subset, we use human-annotated incorrect model predictions from Recht et al. (2019) to select images for which hard negative answer choices can be formed. Starting from 19,056 annotated examples, we narrow the pool to 5,457 images covering 928 of the original 1,000 classes. Because this curation leaves classes unevenly balanced, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A 102-species flower dataset designed around high intra-class variation and strong inter-class similarity. The Flowers test set contains 6,149 examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A 37-category pet dataset with substantial variation in pose, lighting, and occlusion. The Pets test set contains 3,669 examples.

Food-101 (Bossard et al., 2014): A 101-category food dataset with substantial within-class visual diversity. The Food test set contains 25,250 examples.

Question generation. For all fine-grained classification test sets except ImageNet, we generate hard negatives by running zero-shot classification with an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). We take the top 4 incorrect predictions and shuffle them with the correct label to form the answer choices.

Prompt formatting. Our multiple-choice VQA prompt always asks, “What type of object is in this photo?” The answer choices follow the format “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because VLMs are trained with different multiple-choice prompt conventions, we use each model’s corresponding VLMEvalKit format (Duan et al., 2024); for example, some models receive the instruction “Answer with the option’s letter from the given choices directly.”

For CLIP models, we use the standard zero-shot evaluation procedure, so no question prompt is needed. Instead, class names are passed to the text encoder with the template “a photo of a {classname}”.

B.2 GENERAL VQA

For the observational comparison in Section 3, we use the numbers reported on the OpenVLM Leaderboard (Duan et al., 2024). The default leaderboard reports an average score across eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). We use this reported average only in Figure 2.

For all subsequent evaluations, including existing models and the models trained in Section 4, we restrict the general VQA comparison to multiple-choice benchmarks so that the results align more directly with our fine-grained protocol. Specifically, we evaluate VLMs on three benchmarks:

MMBench (Liu et al., 2024b): A 2,948-example benchmark that tests a wide range of VLM capabilities across perception and reasoning. Although some examples are labeled as “fine-grained perception”, they focus on categories such as action recognition, attribute recognition, and OCR, which differ from the fine-grained classification datasets studied here.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal benchmark spanning a broad set of domains and applications. Although the test set contains 10,500 examples, most works report performance on the validation split, which contains 900 questions. MMMU evaluates general knowledge grounded in visual perception across art, science, engineering, medicine, business, and humanities.

MMStar (Chen et al., 2024b): A 1,500-question benchmark selected by human annotators. The examples cover diverse tasks across domains and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks, we use the VLMEvalKit evaluation code (Duan et al., 2024) without modification. Existing models use their predefined evaluation procedures, and our trained models use the LLaVA evaluation code because they are finetuned exclusively on LLaVA-Instruct data.

C TRAINING

We build on the LLaVA-1.5 training codebase (Liu et al., 2023) to keep the ablation platform controlled and reproducible. For most experiments, we retain the LLaVA architecture and training procedure so that changes in performance can be attributed to the targeted ablations.

Pretraining. Pretraining consists of one epoch on either the LLaVA pretraining data, derived from CC-3M images and captions, or Molmo pretraining data (PixMo-Cap). For PixMo-Cap, we use only the “caption” field and exclude the raw audio transcript. During pretraining, the batch size is 256. In most runs, we tune only the randomly initialized MLP connector with a learning rate of $1e-3$. For the pretraining-strategy ablation, we first tune only the connector for a 20% warmup, then unfreeze and tune the LLM with a learning rate of $2e-5$ for the remaining 80% of steps.

Finetuning. For finetuning, every model trains for one epoch on the LLaVA-1.5 finetuning data, which combines LLaVA-Instruct with VQA training data from several datasets to support benchmark evaluation. During finetuning, the LLM learning rate is $2e-5$ and the batch size is 128. The vision encoder remains frozen throughout both training phases.

For the Qwen2-VL architecture ablation, we start from Qwen2-VL-7B Base and apply the same finetuning method, unfreezing only the LLM.

Compute. We train all models on 4 nodes with 4 A100 (40GB) GPUs per node. Training typically finishes within 20 hours, with runtime varying by CLIP model size.

LLaVA vs. PixMo data. The main text contrasts the caption quality of LLaVA and PixMo pretraining data. LLaVA pretraining data comes from CC-3M, a web-scraped caption dataset commonly known to contain low-quality and weakly relevant captions. PixMo-Cap, in contrast, is collected from human annotators who record a 60–90 second audio description of an image, which is then converted into a detailed image caption (Deitke et al., 2024). A surface comparison confirms the difference: LLaVA captions average 9.8 words, whereas PixMo captions average 169 words.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”